PHASE 5 DOCUMENT

EduGenie: Google Gemini Powered Learning Assistant

Phase Name: Quiz Generation, Study Planner and Learning Progress Features

Project Name	EduGenie: Google Gemini Powered Learning Assistant
Phase	Phase 5 – Quiz, Study Planner and Progress
Project Type	AI-Powered Learning Application
Main Technology	Google Gemini API
Prepared By	_____
Guide / Faculty	_____
Date	_____

Academic Project Phase Documentation

1. Phase Overview

This is Phase 5 of **EduGenie: Google Gemini Powered Learning Assistant**. This phase extends the core AI chat experience by adding structured learning tools: quiz generation, study planning, quiz-attempt recording, progress views, and personalized revision support. These features help students move from simply receiving explanations to actively practising, planning, and monitoring their learning.

2. Phase Objective

The objective of this phase is to help students convert learning topics into practical revision activities. EduGenie will generate topic-based quizzes, provide feedback on answers, create study plans based on student goals and available time, and display meaningful learning-progress information. The features are intended to support self-paced preparation while encouraging students to use faculty guidance and approved resources.

3. Quiz Generation Module

The quiz module allows students to select a subject, topic, question type, difficulty level, and number of questions. EduGenie uses structured prompts with Google Gemini to generate practice material such as multiple-choice questions, short-answer questions, true-or-false questions, and scenario-based questions. The system should clearly label AI-generated questions as practice content and should not present them as official examination questions unless approved by faculty.

4. Quiz Attempt and Feedback

Students can answer quiz questions within the application and submit their attempt for evaluation. The system records the selected or entered answers, calculates objective-question scores when an answer key is available, and provides explanations or learning suggestions. For subjective responses, the system can provide formative feedback but should state that the feedback is guidance, not an official academic grade.

5. Study Planner Module

The study planner helps students create a realistic revision schedule. Students can enter the subjects or topics to study, target date, daily available time, current preparation level, and priority areas. EduGenie converts this information into a suggested plan containing daily or weekly tasks, topic order, revision periods, quiz practice, and break reminders. Students can edit the generated plan to match their real schedule.

6. Personalized Revision Support

Based on topics selected by the student and performance in practice quizzes, EduGenie can recommend revision activities such as revisiting a concept, requesting a simpler explanation, attempting another practice quiz, or reviewing saved notes. Recommendations should be presented as optional guidance. The system should not make high-stakes predictions about a student's final academic performance.

7. Learning Progress Dashboard

The progress area displays a student's learning activity in a clear, privacy-aware format. It can show the number of quizzes attempted, topics reviewed, saved learning items, completed study-plan tasks, recent practice scores, and areas marked for revision. The dashboard should make it easy for a student to understand their own learning activity without exposing personal learning data to other students.

8. Data Model Additions

This phase adds QuizSets for generated question collections; QuizQuestions for individual questions and answer keys; QuizAttempts for student responses, scores, and feedback; StudyPlans for generated schedules; StudyTasks for individual plan items and completion status; ProgressSnapshots for summarized learning activity; and LearningRecommendations for optional next-step suggestions. Data should be associated with the correct authenticated student account.

9. Quiz Workflow

1. Student opens the Quiz Generator from the dashboard.
2. Student selects subject, topic, difficulty, question type, and number of questions.
3. EduGenie sends a structured quiz request to the secure Gemini backend.
4. The system displays the generated quiz.
5. Student answers the questions and submits the attempt.
6. EduGenie records the attempt and provides score or formative feedback.
7. The system updates the student's learning-progress view.
8. Student can retry, request an explanation, or save the quiz for later revision.

10. Study Planner Workflow

1. Student opens the Study Planner.
2. Student enters subjects, topics, target date, available study time, and goals.
3. EduGenie validates the details and prepares a planning prompt.
4. Google Gemini returns a suggested study schedule.
5. EduGenie displays the plan as clear tasks or calendar-style items.
6. Student edits, saves, or marks tasks as complete.
7. The progress dashboard updates completed-task information.
8. Student can regenerate or adjust the plan when circumstances change.

11. User Interface Design

The quiz screen should show one question at a time or a clearly organized list, answer controls, progress indication, submit action, and feedback display. The study planner should use simple fields for goals and availability, followed by a readable plan view. The progress dashboard should summarize activity through cards, counts, and optional simple charts without making the interface crowded. All screens should work on mobile devices and remain usable with touch input.

12. Responsible AI and Academic Use

Quiz questions, scores, plans, and recommendations are supportive learning tools. Students should verify important answers with textbooks, instructors, and approved course material. The system should make clear that AI-generated quiz feedback is not an official mark and that a study plan is a suggestion that the student can modify. The application should not use personal learning data for purposes outside the stated educational function.

13. Testing Plan

Testing should check whether quiz parameters are sent correctly, generated questions are displayed properly, answers are recorded against the right student, objective scoring works correctly, feedback appears after submission, study plans are generated from valid input, tasks can be saved and marked complete, progress data updates correctly, and unauthorized users cannot access another student's quiz attempts or study plans. Mobile responsiveness and error conditions should also be tested.

14. Phase Deliverables

The deliverables of this phase are the quiz-generation interface, Gemini-powered quiz prompt templates, quiz attempt and feedback workflow, study planner interface, generated revision schedule, editable study tasks, learning-progress dashboard, supporting database entities, responsible-use notices, and test results for the new learning features.

15. Planned Next Phase

The next phase will be **Phase 6: Project Assistance, Faculty Support, and Resource Management**. It will add project-oriented guidance, documentation support, faculty resource sharing, approved learning materials, and features that connect student learning assistance with faculty-led academic activities.

16. Conclusion

Phase 5 strengthens EduGenie by transforming AI explanations into active learning support. Quiz generation gives students a way to practise, study planning helps them organize revision, and progress features help them reflect on their own learning activity. Together, these tools promote structured, independent, and responsible learning.

Phase Approval

Prepared by	_____
Guide / Faculty	_____
Approved by	_____